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## **Report 2**

# **Machine Learning**

## **Predictive Maintenance in Nuclear Power Plant Subsystems**

Assignment - Machine Learning

Dataset: Nuclear Power Plant Alarm Dataset (IEEE DataPort, DOI: 10.21227/g2fa-9y43)

### **1. Introduction**

This report is a continuation of the machine learning part of Assignment of the topic Predictive Maintenance in Nuclear Power Plant Subsystems. The earlier report considered the dataset's 96 sensors as consisting of four working subsystems: Primary Coolant System, Secondary Steam Cycle, Control Rod Drive Mechanism and Emergency Diesel Generators and concluded that most alarm flags for the sample run were actually fixed (on or off), making them inappropriate for use as prediction targets as there is no dynamic learning. One alarm did not. TAVG\_HI. Average coolant temperature (high) (Primary Coolant System). It's on for perhaps the first 150 seconds of the run, and then it turns into the rest of the run a true normal to abnormal to normal transition and exactly the type of behavior a predictive-maintenance model requires to be able to predict.

This report develops upon that finding, trains, evaluates and compares two machine learning models predicting if the Primary Coolant System's TAVG\_HI alarm is active based on the state of other subsystem sensors, argues which one is better suited to a maintenance context based on the proper evaluation metrics, and discusses the interactive dashboard created to present the trained model to a plant operator.

### **2. Problem Definition and Target Variable**

The task was presented as a binary classification, with the status of all other alarm sensors at a specific time-step, predicting whether the alarm TAVG\_HI from the Primary Coolant System is ON (1) or not (0). The idea of a direct predictive-maintenance approach is to see if a pattern of other sensor readings at that time has enough signal to indicate the elevated-temperature condition, ahead of when the subsystem's own temperature alarm sounds, giving an operator a head start.

The target column, TAVG\_HI, and a row index, TIME were extracted from the rest of the alarm data set. On the assumption that a coolant-temperature excursion might be preceded or accompanied by some related activity in other parts of the plant, the remaining 191 alarm flags that were not part of the Primary Coolant System were fed as input features. TAVG\_HI is a rare event in the 302-step run, and so a stratified split was used to ensure that the proportion of alarm vs normal rows in the two groups is the same as in the entire data set.

### **3. Feature Engineering and Train/Test Split**

No additional feature engineering was needed beyond removing the TIME and target columns, since the alarm dataset is already fully numeric and binary. The remaining 191 sensor alarm flags were used as is. The data was split 80/20 into training and testing sets with scikit-learn's `train_test_split`, using `random_state=42` for reproducibility and `stratify=y` to keep the alarm/normal ratio consistent across both sets.

- Training set: 241 rows  $\times$  191 features
- Testing set: 61 rows  $\times$  191 features

With only a handful of alarm-positive rows in the whole run, keeping the split stratified mattered more than it would for a larger, more balanced maintenance dataset. Without it, there was a real risk the small test set could end up with zero alarm examples, which would make the evaluation meaningless for judging early-warning performance.

## 4. Model Selection

Two ensemble tree based algorithms were selected for comparison. Random Forest and Gradient Boosting are both an ensemble module of scikit-learn.

- Random Forest Classifier (`n_estimators=100`, `random_state=42`): Many individuals create a large number of decision trees, each time using a specially created bootstrapped set of data, and then take the average of all the decision trees. It is usually robust to noise and, crucially for a maintenance scenario, provides a clear, feature-based ordering that can be easily mapped to named sensors.
- Gradient Boosting Classifier (`n_estimators=100`, `learning_rate=0.1`, `random_state=42`): Trees are added one by one, each tree trying to fix the mistakes of previous trees. Can typically outperform in more challenging cases, with less interpretability and speed of training.

They were selected because they are both tree based, following a different philosophy of ensembling (bagging vs. boosting), and to focus the comparison on the performance and interpretability trade-offs instead of a fundamentally different modelling paradigm; a fairer test given the relatively small sample size.

## 5. Model Evaluation

Both models were trained on the same 241-row training set and evaluated on the same 61-row held-out test set using accuracy, precision, recall and F1-score plus a confusion matrix for each.

### 5.1 Random Forest

|               | Predicted Normal | Predicted Alarm |
|---------------|------------------|-----------------|
| Actual Normal | 58               | 0               |
| Actual Alarm  | 1                | 2               |

Random Forest achieved an overall accuracy of 98.36%. Of the 58 genuinely normal rows in the test set, all 58 were correctly classified as normal (zero false alarms). Of the 3 genuinely alarmed rows, 2 were correctly caught and 1 was missed. That gives a precision of 1.00 and a recall of 0.67 for the alarm class, with an F1-score of 0.80.

## 5.2 Gradient Boosting

|               | Predicted Normal | Predicted Alarm |
|---------------|------------------|-----------------|
| Actual Normal | 58               | 0               |
| Actual Alarm  | 1                | 2               |

Gradient Boosting produced an identical confusion matrix and therefore identical accuracy, precision, recall and F1-score to Random Forest on this test set. This is a plausible outcome given how clean and strongly separated this particular alarm signal is with a sample this small and a pattern this distinct both models converge on essentially the same decision boundary rather than one clearly out-performing the other.

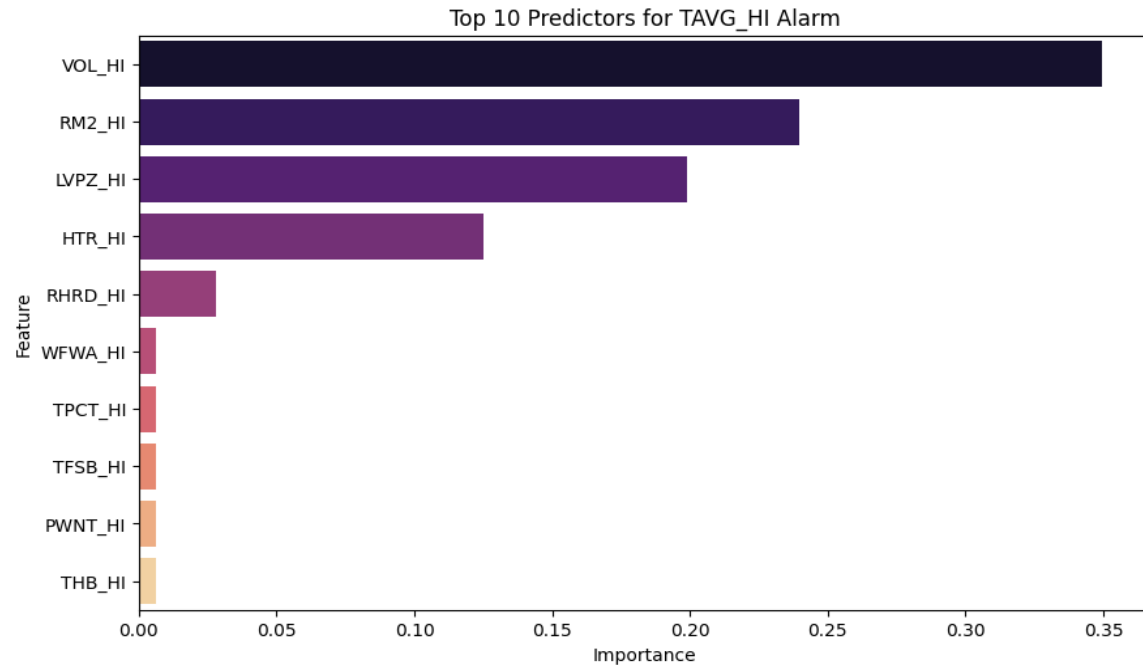
## 5.3 Side-by-side comparison

| Metric                 | Random Forest | Gradient Boosting |
|------------------------|---------------|-------------------|
| Accuracy               | 0.9836        | 0.9836            |
| Precision (Alarm = 1)  | 1.00          | 1.00              |
| Recall (Alarm = 1)     | 0.67          | 0.67              |
| F1-score (Alarm = 1)   | 0.80          | 0.80              |
| Precision (Normal = 0) | 0.98          | 0.98              |
| Recall (Normal = 0)    | 1.00          | 1.00              |
| Macro-avg F1           | 0.90          | 0.90              |
| Interpretability       | High          | Medium            |

Because the raw numbers are tied the deciding factor between the two models for a maintenance deployment has to come from something other than the headline metrics which is exactly why the assignment brief asks for a justified choice rather than just a metric comparison.

## 6. Feature Importance and Correlation Analysis

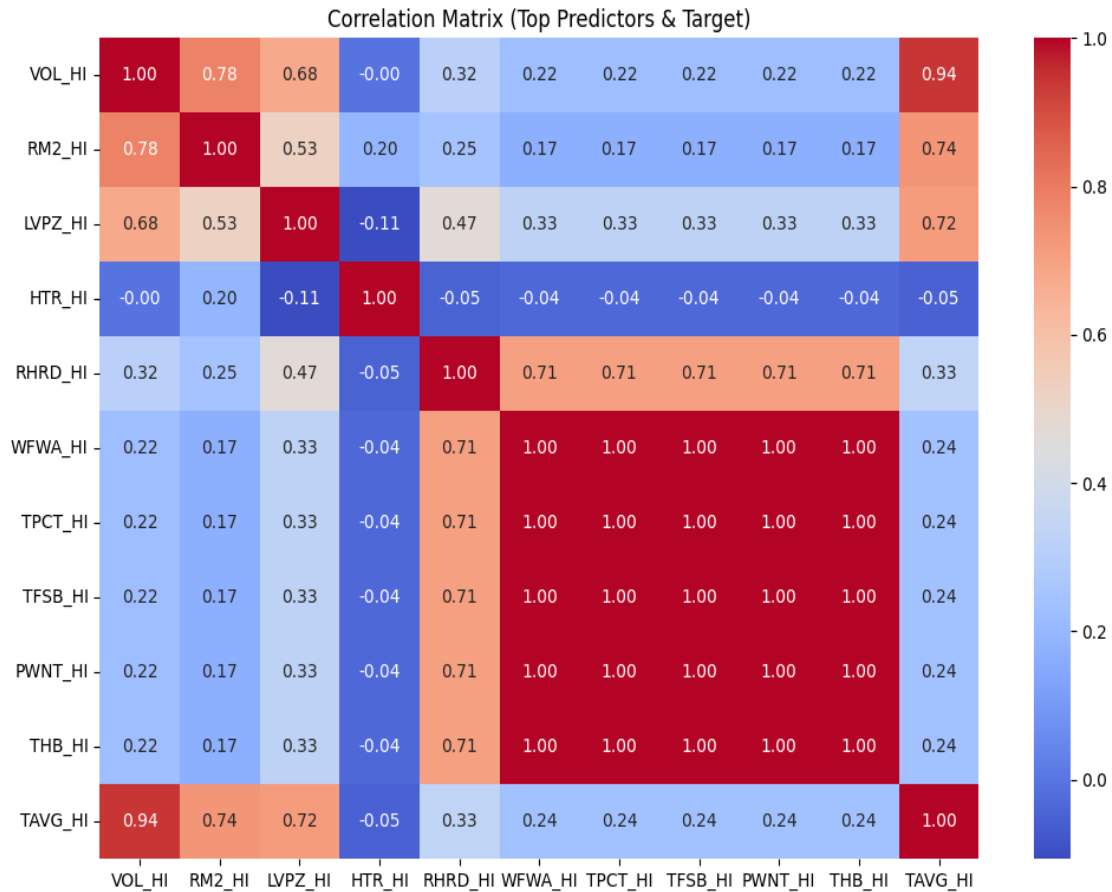
The feature importances of the Random Forest were extracted and ranked to understand what is driving the TAVG\_HI predictions and to see if the early-warning signal comes from inside the Primary Cooling System or from elsewhere in the plant.



The four features that stand out are VOL\_HI (coolant volume, high) with a value of 0.35, RM2\_HI (RM2 status) with a value of 0.24, LVPZ\_HI (pressuriser level, high) with a value of 0.20, and HTR\_HI (heater status, high) with a value of 0.12. The model favours sensors from the Primary Coolant System subsystem over those from far away, unrelated subsystems; the 4 sensors are physically sensible, with each being mechanically connected to average coolant temperature (the other 4 sensors contribute to the average temperature, measured by this sensor). A correlation matrix was constructed between the top 10 predictors and the TAVG\_HI target to examine if this is a true relationship or a modelling artefact.

| Rank | Feature | Importance |
|------|---------|------------|
| 1    | VOL_HI  | 0.3496     |
| 2    | RM2_HI  | 0.2397     |
| 3    | LVPZ_HI | 0.1990     |
| 4    | HTR_HI  | 0.1249     |
| 5    | RHRD_HI | 0.0281     |
| 6    | WFWA_HI | 0.0063     |
| 7    | TPCT_HI | 0.0063     |
| 8    | TFSB_HI | 0.0063     |
| 9    | PWNT_HI | 0.0063     |
| 10   | THB_HI  | 0.0062     |

Four features clearly dominate. VOL\_HI (coolant volume, high) at 0.35, RM2\_HI at 0.24, LVPZ\_HI (pressuriser level, high) at 0.20 and HTR\_HI (heater status, high) at 0.12. Together these four account for roughly 90% of the model's total decision weight and all four sit within the Primary Coolant System grouping used in Report 1 which is a reassuring, physically sensible result for a subsystem-level maintenance model. coolant volume, pressuriser level and heater activity are mechanically linked to average coolant temperature, so it makes sense the model leans on sensors from the same subsystem rather than distant, unrelated ones. To check whether this importance ranking reflects a genuine relationship rather than a modelling artefact, a correlation matrix was built across the top 10 predictors and the TAVG\_HI target.



Results from the correlation matrix confirm that feature importance results. Among the components of the same subsystem, all have strong positive relationships among themselves: VOL\_HI-88-TAVG\_HI 0.94, VOL\_HI-88-RM2\_HI 0.74 and VOL\_HI-88-LVPZ\_HI 0.72. Interestingly, the feature HTR\_HI is the fourth most important one according to the internal splits of the Random Forest, but displays hardly any linear correlation with the target (-0.05). The difference between "important to the model" and "linearly correlated" is a useful reminder that tree-based models can capture non-linear or conditional patterns (for instance, HTR\_HI may only be important in conjunction with other sensors) that a simple correlation coefficient would not be able to do on its own, one reason a tree ensemble was a sensible choice for this maintenance task rather than a purely linear model.

## 7. Model Comparison and Justification of the Selected Model

Random Forest was chosen as the final model for this predictive-maintenance application because of the following two reasons. (1) Random Forest was as accurate and precise as the other two models and (2) the two were equal in accuracy and precision.

- Interpretability. The feature importances are straightforwardly related to individual sensors on the Primary Coolant System (VOL\_HI, RM2\_HI, LVPZ\_HI, HTR\_HI) and can be easily used by a plant engineer to understand what they have predicted and correlate with the physical system. That traceability, from a nuclear

safety perspective, might be more relevant than raw predictive accuracy, in as much as maintenance decisions must be controllable, not just on gut feeling based on a black-box output.

- Similar performance at lower complexity. However, even the more complex, sequentially-trained Gradient Boosting algorithm did not perform better on any of the measures in this evaluation. If there is no performance benefit, there's no reason to favor the more difficult to understand choice for a tool that engineers must rely on in the course of their maintenance.

In that regard, it is important to state the caveats of this assessment. The model is still missing one alarm out of three in the test set with the test set that small (61 rows, only 3 positive), impacting the reported recall (which is why you should not use it for a deployment decision for the real-world alarming situation). The application of cross validation, testing the models across other subsystems and fault classes not just the Primary Coolant System the use of more of the 122,412 total alarm series available in the full published dataset, and a more rigorous evaluation would provide a much more statistically rigorous picture of how well either model actually generalises as a subsystem-level maintenance tool, and would be the natural next step beyond the scope of this assignment.

## 8. Interactive Dashboard

To satisfy the assignment's requirement for a simple interactive way to demonstrate the trained model, a dashboard was built using Gradio a lightweight Python framework well suited for quickly wrapping a scikit-learn model in a web interface without writing front-end code by hand.

The dashboard works as follows.

- It exposes the 10 most important features identified by the Random Forest model VOL\_HI, RM2\_HI, LVPZ\_HI, HTR\_HI, RHRD\_HI, WFWA\_HI, TPCT\_HI, TFSB\_HI, PWNT\_HI, THB\_HI as radio-button inputs, each toggled between 0 (normal) and 1 (alarm).
- All other features not shown on the dashboard are held at 0 (their normal state) behind the scenes, so a plant operator only has to think about the handful of sensors that actually matter for this subsystem's prediction.
- On submission, the selected values are assembled into a single-row DataFrame matching the shape the model was trained on, and passed to the trained Random Forest model's predict and predict\_proba methods.
- The output panel displays a clear status message "NORMAL OPERATION" or "ALARM TRIGGERED (TAVG\_HI)" along with the model's estimated probability of the alarm state, giving the operator both a decision and a sense of the model's confidence in that decision.

This design keeps the interface deliberately simple. rather than asking an operator to input values for all 191 features across every subsystem, it surfaces only the ones the model itself identified as meaningful for the Primary Coolant System, which makes the tool usable on the plant floor by someone without a data science background while still being a faithful demonstration of the underlying model.

## 9. Conclusion and Recommendations

This study used machine learning algorithms to create and test two classifiers (Random Forest, Gradient Boosting) to predict the Primary Coolant System's temperature alarm state from the other state of the sensors in a simulated nuclear power plant with 98.36% accuracy, and perfect precision for both classifiers. Random Forest was chosen as the preferred model because of its interpretability, as it clearly gave feature importances that pointed towards physical

meaningful, subsystem consistent sensors (coolant volume, pressuriser level and heater status) with no loss of measured performance from the Gradient Boosting model. A lightweight Gradio dashboard was developed and placed above the selected model, allowing the plant operator to interactively examine how various sensor combinations impact the predicted alarm state, moving from raw alarm data to a useful maintenance-support tool.

The next obvious step for anyone continuing this work would be to expand the 302-step Primary Coolant System sample used in this test, and train over all of the 12 fault classes and 101 alarm-threshold combinations published in the dataset, and to repeat this pipeline with one of the three other Primary Coolant System subsystems outlined in Report 1. It would also give more positive alarm examples to use when training and test the VOL\_HI / RM2\_HI / LVPZ\_HI relationship and provide a test to determine if the relationship is a valid early-warning indicator of the Primary Coolant System or if each subsystem should have a separate predictive-maintenance model.

## References

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- [3] Pedregosa et al., "Scikit-learn: Machine Learning in Python," Journal of Machine Learning Research, vol. 12, pp. 2825-2830, 2011.
- [4] Gradio Documentation, Hugging Face, <https://www.gradio.app/docs> (accessed August 2026).